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CoE-XII

Aakash
Medical | IIT-JEE | Foundations
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NCERT BOOSTER TEST SERIES

Test No. –03

Topics covered in various subjects:

Physics : System of particles and Rotational motion, Gravitation, Mechanical Properties of solids

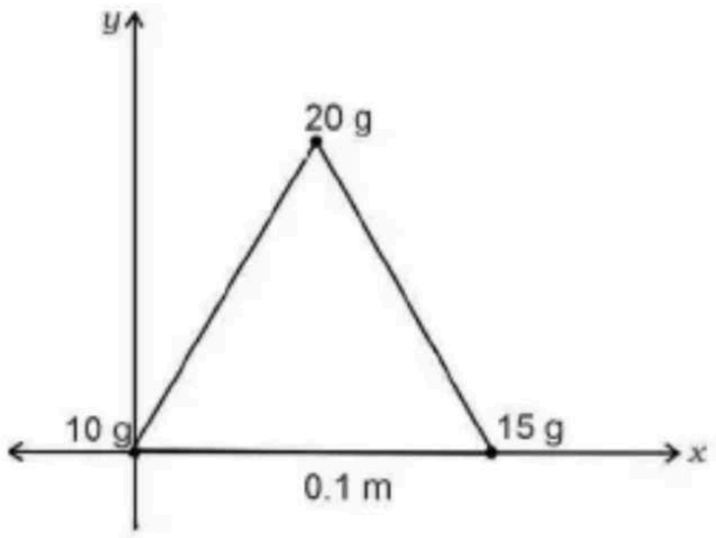
Chemistry: Equilibrium, Redox reactions

Botany : Cell Cycle and Cell Division, Transport in Plants

Zoology : Neural Control and Coordination

Choose the correct answer:

1. Three particles of mass 10 g, 15 g and 20 g are placed at the vertices of equilateral triangle of side of 0.1 m as shown. Centre of mass will be located at the point



(1) $\left(\frac{5}{18}, \frac{1}{3\sqrt{3}}\right)m$ (2) $\left(\frac{1}{3\sqrt{3}}, \frac{15}{18}\right)m$

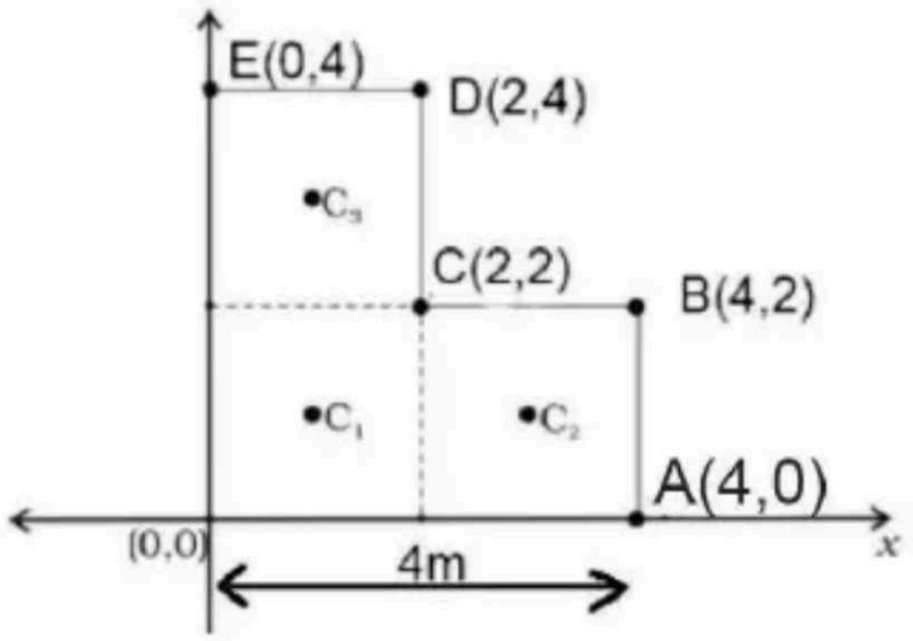
(3) $\left(\frac{1}{15\sqrt{3}}, \frac{1}{18}\right)m$ (4) $\left(\frac{1}{18}, \frac{1}{15\sqrt{3}}\right)m$

2. Mass is distributed uniformly over a thin triangular lamina. If co-ordinates of vertices are (– 6, 2)m, (2,6)m and (4, – 8) m then the Co-ordinates of centre of mass is

(1) (0,0) (2) $(-3,2)m$

(3) $\left(\frac{3}{2}, \frac{5}{3}\right)m$ (4) $\left(-\frac{7}{2}, \frac{4}{7}\right)m$

3. A uniform L-shaped lamina (a thin flat plate) with dimensions as shown. If the mass of the lamina is 6kg then the centre of mass of lamina is



(1) $\left(\frac{5}{6}m, \frac{5}{6}m\right)$ (2) $\left(\frac{5}{2}m, \frac{5}{2}m\right)$

(3) $\left(\frac{5}{2}m, \frac{5}{6}m\right)$ (4) $\left(\frac{5}{3}m, \frac{5}{3}m\right)$

4. A projectile, following the usual parabolic trajectory, explodes into fragments midway in air. The centre of mass of fragments will follow

(1) Horizontal rectilinear path

(2) Vertical rectilinear path

(3) Any parabolic path

(4) Same parabolic path

5. The motion of centre of mass, of a system of particles depends

(1) On the external forces

(2) On the internal forces

(3) On the nature of the system of particles

(4) All of these

6. A boy sits stationary at one end of a long trolley moving uniformly with a speed 20 km/h on a smooth horizontal floor. If the boy gets up and runs with a speed 3km/h on the trolley in the same direction, the speed of the CM of the (trolley + boy) system is

(1) 20 km/h (2) 3 km/h

(3) 23 km/h (4) 17 km/h

7. Select incorrect option about the vector product

(1) The scalar product of two vectors is commutative

(2) The scalar product of two vectors is distributive with respect to vector condition.

(3) The vector product of two vectors is commutative

(4) The vector product of two vectors is distributive with respect to vector addition

8. The vector products of two vectors, $\vec{a}=\left(3\hat{i}-4\hat{j}+2\hat{k}\right)$ and $\vec{b}=\left(-\hat{i}+2\hat{j}-\hat{k}\right)$, is

(1) $\hat{i}-\hat{j}+2\hat{K}$ (2) $2\hat{i}+\hat{j}-2\hat{k}$

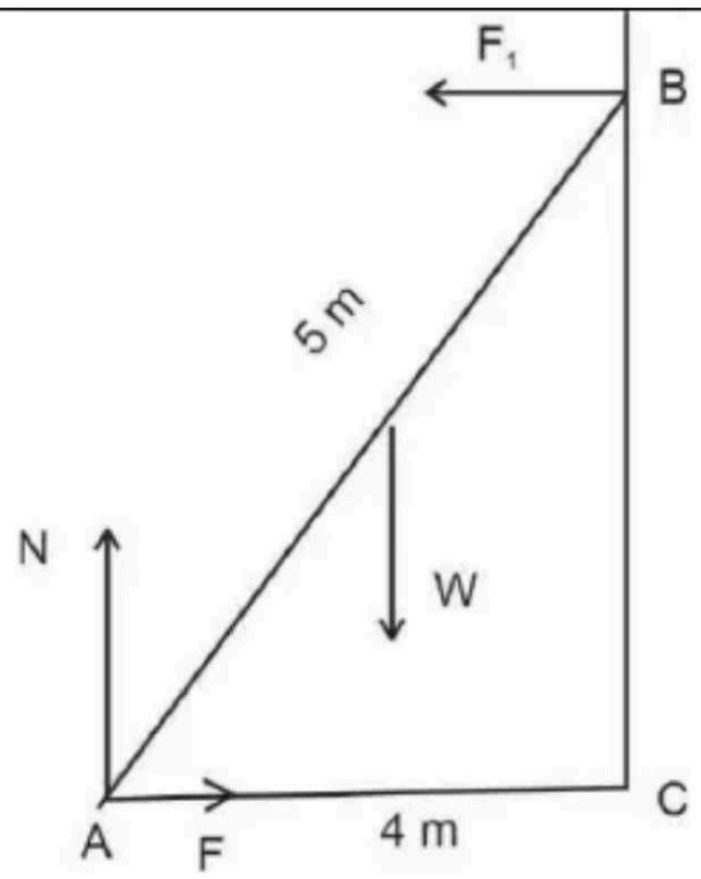
(3) $-\hat{j}+2\hat{k}$ (4) $\hat{j}+2\hat{k}$

9. The magnitude of the area of the triangle contained between the vector $\vec{a}=2\hat{i}+\hat{j}+\hat{k}$ and $\vec{b}=-\hat{i}+3\hat{j}+4\hat{k}$, is

(1) 131 unit (2) $\frac{\sqrt{131}}{2}$ unit

(3) $\sqrt{\frac{131}{2}}$ Unit (4) $\sqrt{131}$ unit

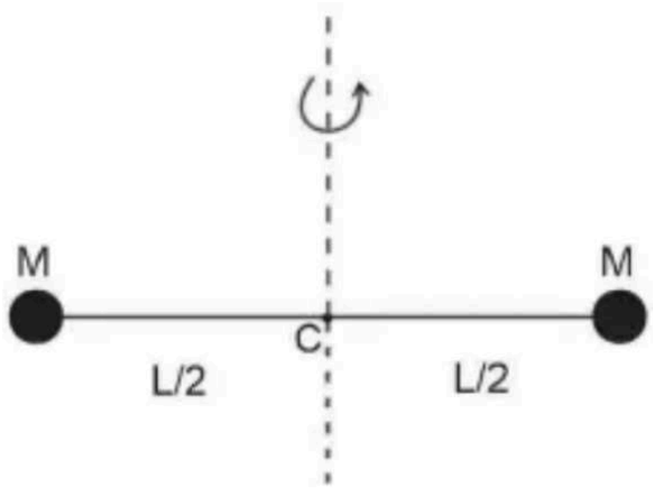
10. A 5m long ladder weighing 10 kg leans on a frictionless wall. Its feet rest on the floor 4 m from the wall. The reaction force of the wall is



(1) 98 N (2) 200 N

(3) 65.3 N (4) 76 N

11. A rigid rod of negligible mass of length L with a pair of small masses, rotating about an axis through the centre of mass perpendicular to the rod as shown in the figure. The moment of inertia of the masses is



(1) $\frac{ML^2}{2}$ (2) $\frac{ML^2}{4}$

(2) ML^2 (4) $2ML^2$

12. The radius of gyration of the circular disc of radius R about its diameter is

(1) R (2) $\frac{R}{\sqrt{2}}$

(3) $\frac{R}{2}$ (4) $\frac{R}{2\sqrt{2}}$

13. A rope of negligible mass is wound round a hollow cylinder of mass 2 Kg and radius 20 cm. If the rope is pulled with a force of 10 N then the linear acceleration of the rope is

(1) 10 m/s^2 (2) 25 m/s^2

(3) 150 m/s^2 (4) 5 m/s^2

14. A hoop of radius 1 m weighs 50 kg. It is rolling without slipping along a horizontal floor so that its centre of mass has a speed 10 cm/s. The work done to stop it, is

(1) 4J (2) 0.5 J

(3) 2J (4) 1J

15. A solid cylinder rolls up an inclined plane of angle of inclination 37°. At the bottom of the inclined plane the centre of mass of the cylinder has a speed of 4m/s. The maximum distance covered by the cylinder on the plane is

(1) 3.8 m (2) 2 m

(3) 4.6 m (4) 3 m

16. A solid cylinder of mass 10kg rotates about its axis with angular speed 50 rad s^{-1}. The radius of the cylinder is 0.1 m. The magnitude of angular momentum of the cylinder about its axis is

(1) 6.25 Js (2) 2.5 Js

(3) 10 Js (4) 5 Js

17. A child stands at the centre of a turntable with his two arms outstretched. The turntable is set rotating with an angular speed of 60 rev/min If he folds his hands back and thereby reduces his moment of inertia to $\frac{1}{4}$ times the initial value then the angular speed of child is

(1) 8 rev/min (2) 240 rev/min

(3) 4 rev/min (4) 100 rev/min

18. To maintain a rotor at a uniform angular speed of 100 $rads^{-1}$, an engine needs to transmit a torque of 80 Nm. If the efficiency of engine is 100%, then the power required by the engine is

(1) 8 kW (2) 18 kW

(2) 9 kW (4) 36 kW

19. A rigid body is in mechanical equilibrium if

(1) It is in translational equilibrium only

(2) It is in rotational equilibrium only

(3) It is in translational as well as rotational equilibrium

(4) It is neither in translational nor in rotational equilibrium

20. The centre of gravity of an extended body is that point where

(1) The total mass of the body is assumed to be concentrated

(2) The total torque on the body is zero

(3) The total gravitational torque on the body is zero

(4) The total force is zero

21. According to Kepler, the law of area can be understood as a consequence of conservation of

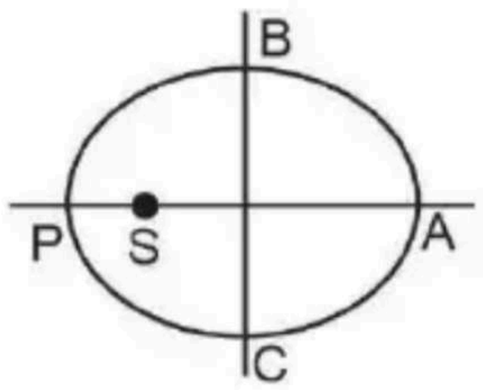
(1) Linear momentum

(2) Angular momentum

(3) Mass

(4) Energy

22. The figure shows an elliptical orbit of a planet about the sun S. If t_1 is the time for planet to traverse BAC and t_2 is the time to traverse CPB then



(1) $t_1 = 2t_2$ (2) $t_1 < t_2$

(3) $t_1 > t_2$ (4) $t_1 = t_2$

23. In considering the motion of an object under the gravitational influence of another object a physical quantity which is not conserved, is

(1) Angular momentum

(2) Total mechanical Energy

(3) Linear momentum

(4) All of these

24. An astronaut experiences weightlessness in a space satellite. It is because

(1) Only astronaut is in 'free fall' towards the earth

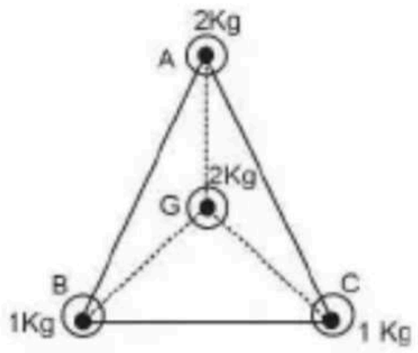
(2) Only satellite is in "free fall" towards the earth

(3) Astronaut and satellite (both) are in "free fall" towards the earth

(4) Gravitational force is small at that location in space

25. Three masses of 1Kg, 1Kg, and 2Kg are placed at the vertices of an equilateral triangle ABC as shown. The force acting on a mass 2 kg placed at the centroid G of the triangle is

[Take AG = BG = CG = 1m]



(1) 6.67×10^{-11} N (2) 13.34×10^{-11} N

(3) 26.68×10^{-11} N (4) Zero

26. A satellite of mass *m* is in a circular orbit of radius 2R about the Earth. The energy required to transfer it to a circular orbit of radius 4 R is [R is radius of Earth]

(1) $\frac{mgR}{8}$ (2) mgR

(3) $\frac{mgR}{4}$ (4) $\frac{mgR}{2}$

27. A body weighs 54 N on the surface of the Earth. The gravitational force on it due to earth at a height equal to half of the radius of the earth is

(1) 63 N (2) 45 N

(3) 24 N (4) 20 N

28. The escape speed of a body from the earth does not depend on

(1) The mass of the body

(2) The location from where it is projected

(3) The direction of projection

(4) All of these

29. The escape speed of a projectile on the earth's surface is 11.2 kms^{-1}. A body is projected with twice this speed. The speed of the body far away from the earth is nearly

(1) $16kms^{-1}$ (2) $20kms^{-1}$

(3) $15kms^{-1}$ (4) $19kms^{-1}$

30. The most accurate expression for the gravitational potential energy associated with two particles of masses m_1 and m_2, separated by a distance r, is

(1) $U = -\frac{Gm_1m_2}{r}$

(2) $U = -\frac{Gm_1m_2}{2r}$

(3) $U = -\frac{Gm_1m_2}{r} + K(\text{constant})$

(4) $U = -\frac{Gm_1m_2}{2r} + k(\text{constant})$

31. The most accurate formula for the difference of potential energy between two points r_2 and r_1 distance away from the centre of earth is

[Symbols have their usual meaning]

(1) $-GMm\left[\frac{1}{r_2}-\frac{1}{r_1}\right]$

(2) $mg[r_2 - r_1]$

(3) $-GMm\left[\frac{1}{r_2}-\frac{1}{r_1}\right]+K$ (constant)

(4) $mg[r_2 - r_1]+K$ (constant)

32. If potential energy is assumed to be zero at infinity then

(1) The total energy of an orbiting satellite is negative of its potential energy.

(2) The total energy of an orbiting satellite is twice of its potential energy

(3) The total energy of an orbiting satellite is negative of its kinetic energy

(4) The total energy of an orbiting satellite is twice of its kinetic energy

33. The energy required to launch an orbiting satellite out of earth's gravitational influence is

(1) More than the energy required to project a stationary object at the same height (as the satellite) out of earth's influence

(2) Less than the energy required to project a stationary object at the same height (as the satellite) out of earth's influence

(3) Twice the energy required to project a stationary object at the same height (as the satellite) out of earth's influence

(4) Zero

34. Assuming the earth to be a sphere of uniform mass density if a body weighed 200 N on the surface of the earth thus the weight of it at half way down to the centre of earth is

(1) 50 N (2) 100 N

(3) 40 N (4) 20 N

35. Two spheres, each of mass 10 kg and radius 0.20 m, are placed 2 m apart on a horizontal table. The gravitational potential at mid-point of the line joining the centres of the spheres, is

(1) -2.7×10^{-8} J/kg

(2) -2.4×10^{-7} J/kg

(3) -3.2×10^{-9} J/kg

(4) -1.3×10^{-9} J/kg

36. Bulk modulus is defined for

(1) Solids only

(2) Liquids only

(3) Gases only

(4) Solids, liquids and gases

37. A structural steel rod has a radius of 20 mm and a length of 2.0 m. If a 10 kN force stretches it along its length then stress produced in it is nearly

(1) 3×10^8 Nm^{-2} (2) 3×10^6 Nm^{-2}

(3) 6×10^8 Nm^{-2} (4) 8×10^6 Nm^{-2}

38. A steel wire of length 4 m and cross-section area 2×10^{-5} m^2 stretches by the same amount as a copper wire of length 2.5 m and cross-section area of 4×10^{-5} m^2 under a given load. The ratio of Young's modulus of steel to that of copper is

(1) 1.2 (2) 1.8

(3) 2.5 (4) 3.2

39. The average depth of an ocean is about 2200 m, if the bulk modulus of water is 2.2×10^9 Nm^{-2}, then the fractional compression $\frac{\Delta V}{V}$ of water is

(1) 0.2 (2) 0.02

(3) 0.1 (4) 0.01

40. Which of the following substances has the highest elasticity?

(1) Aluminium (2) Brass

(3) Copper (4) Steel

41. In case of bending of a beam, depression δ depends on length of beam (L) as

(1) $\delta \propto \frac{1}{L}$ (2) $\delta \propto L^2$

(3) $\delta \propto \frac{1}{L^3}$ (4) $\delta \propto L^3$

42. A steel cable with a radius of 1 cm supports a chairlift at a ski area. If the maximum stress is not to exceed 10^8 N/m^2, the maximum load, the cable can support is

(1) 7×10^{-4} N (2) 6.28×10^4 N

(3) 3.14×10^4 N (4) 31.4×10^4 N

43. A glass slab is subjected to a hydraulic pressure of 5 atm, the fractional change in volume of glass slab is

[Bulk modulus for glass = 37 GPa]

(1) 3.12×10^{-8} (2) 1.35×10^{-5}

(3) 2.21×10^{-6} (4) 5.25×10^{-7}

44. A solid copper cube of an edge 20 cm is subjected to a hydraulic pressure of 14×10^6 Pa. If Bulk modulus for copper is 140 GPa, then the change in volume of the cube is

(1) 8×10^{-7} m^3 (2) 5×10^{-7} m^3

(3) 5×10^{-8} m^3 (4) 6×10^{-8} m^3

45. The change in the pressure on a litre of water to compress it by 0.5% is

[Bulk modulus of water = 2.2 GPa]

(1) 11×10^6 Pa (2) 2.2×10^6 Pa

(3) 22×10^6 Pa (4) 20×10^6 Pa

Chemistry

46. Select the incorrect statement

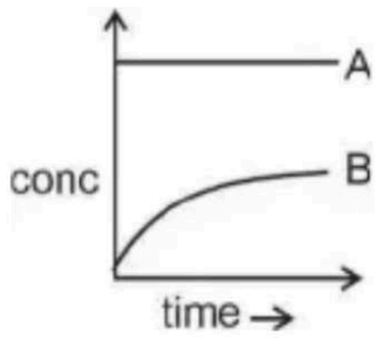
(1) Both the opposing processes stop at equilibrium

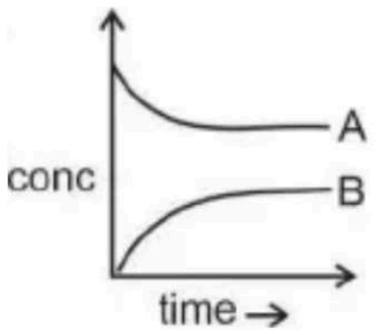
(2) Equilibrium is possible only in a closed container

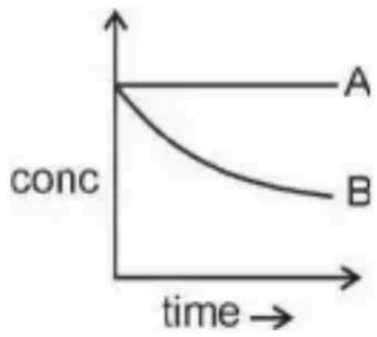
(3) All the measurable properties of the system remain constant at equilibrium

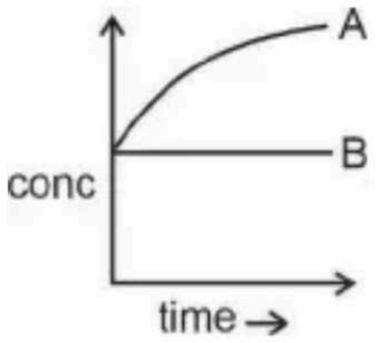
(4) There is dynamic but stable condition at equilibrium

47. For a gaseous equilibrium A(g) $\rightleftharpoons$ B(g) which of the following graph is correct?

(1) 

(2) 

(3) 

(4) 

48. Expression of equilibrium constant for the reaction $\frac{1}{2}N_2(g)+\frac{3}{2}H_2(g) \rightleftharpoons NH_3(g)$ is

(1) $\frac{[NH_3]^2}{[N_2][H_2]^3}$ (2) $\frac{[N_2][H_2]^3}{[NH_3]^2}$

(3) $\frac{[NH_3]}{[N_2]^{1/2}[H_2]^{3/2}}$ (4) $\frac{[N_2]^{1/2}[H_2]^{3/2}}{[NH_3]}$

49. If equilibrium constant for the reaction $2A+3B \rightleftharpoons 2C$ is K then equilibrium constant for the reaction $C \rightleftharpoons A+\frac{3}{2}B$ is

(1) $\frac{1}{K}$ (2) $\frac{1}{K^2}$

(3) $\frac{1}{\sqrt{K}}$ (4) K^2

50. K_P is greater than K_C for the reaction

(1) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

(2) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

(3) $N_2O_4(g) \rightleftharpoons 2NO_2(g)$

(4) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

51. If initially 3M each of CO and H_2O are present and equilibrium constant $K_C = 4$ the equilibrium concentration of CO_2 gas for the reaction $CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$ is

(1) 1 M (2) 2 M

(3) 3M (4) 4M

52. Heterogeneous equilibrium among the following is

(1) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

(2) $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

(3) $2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$

(4) $CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$

53. Unit of K_p for the reaction $2A(g) + 2B(g) \rightleftharpoons 3C(g)$ is

(1) atm (2) atm^{-1}

(3) atm^3 (4) atm^{-3}

54. If $K_C > 10^3$ then

(1) Reactants predominate over products

(2) Both reactants and products are present at equilibrium in appreciable amount

(3) Reaction hardly proceeds

(4) Products predominate over reactants

55. Select the incorrect statement

(1) If $Q_C < K_C$, net reaction goes from left to right

(2) If $Q_C > K_C$, net reaction goes from right to left

(3) If $Q_C = K_C$, individual reactions stop

(4) If $Q_C = K_C$, then reaction is at equilibrium

56. Correct relation among the following

(1) $\Delta G^\circ = \Delta G + RT \ln Q$ (2) $\Delta G = \Delta G^\circ + RT \ln Q$

(3) $\Delta G^\circ = RT \ln K$ (4) $\Delta G = RT \ln k$

57. Which of the following condition will shift the equilibrium

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ in forward direction

(1) Adding PCl_3 (2) Removing PCl_5

(3) Increasing volume (4) Increasing pressure

58. For the reaction

$2NO_2(g) \rightleftharpoons N_2O_4(g)$; $\Delta H = -57.2$ kJ mol^{-1}

The value of K_C can be increased by

(1) Adding NO_2

(2) Adding N_2O_4

(3) Increasing temperature

(4) Decreasing temperature

59. Addition of inert gas at constant pressure exert no effect on

(1) $H_2(g) + Cl_2(g) \rightleftharpoons 2HCl(g)$

(2) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

(3) $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

(4) $C(s) + CO_2(g) \rightleftharpoons 2CO(g)$

60. Conjugate base of $H_2PO_4^-$ is

(1) H_3PO_4 (2) HPO_4^{2-}

(3) PO_4^{3-} (4) $H_2PO_3^-$

61. Which of the following ions is/are possible

(1) $H_3O_2^-$ (2) $H_5O_3^-$

(3) H_3O^+ (4) All of these

62. Which of the following is not an Arrhenius acid

(1) H_3O^+ (2) HCl

(3) BF_3 (4) HSO_4^-

63. Molarity of pure water in nearly

(1) 1 M (2) 18 M

(3) 55.5 M (4) 10^{-7} M

64. Weakest base among the following is

(1) H^- (2) NH_2^-

(3) O^{2-} (4) H_2O

65. pH of 10^{-10} M HCl aqueous solution is nearly

(1) 2 (2) 4

(3) 7 (4) 10

66. Which of the following fluid has pH < 7

(1) Egg white (2) Sea water

(3) Human blood (4) Human saliva

67. pH of 10^{-1} M aqueous solution of CH_3COOH is (pKa CH_3COOH = 4.7)

(1) 1.4 (2) 2.85

(3) 3.7 (4) 4.7

68. If K_a of NH_4^+ is x then K_b of NH_3 is

(1) $\frac{K_w}{x}$ (2) $K_w.x$

(3) $\frac{x}{K_w}$ (4) K_w

69. For citric acid, which of the following option is correct?

(1) $K_{a_1} < K_{a_2} < K_{a_3}$ (2) $K_{a_1} > K_{a_2} > K_{a_3}$

(3) $K_{a_1} = K_{a_2} = K_{a_3}$ (4) $K_{a_1} > K_{a_2} < K_{a_3}$

70. Common ion effect is not observed in

(1) NaCl + HCl

(2) CH_3COOH + HCl

(3) NH_4OH + NH_4Cl

(4) CH_3COOH + CH_3COONa

71. Which of the following ion shows hydrolysis

(1) NO_3^- (2) Cl^-

(3) CH_3COO^- (4) Br^-

72. If pK_a of weak acid HA is 4 and pK_b of weak base BOH is 5 then pH of aqueous solution of BA is

(1) 7.5 (2) 6.5

(3) 4 (4) 5

73. pH of equimolar aqueous solution of CH_3COONa and CH_3COOH is (pK_a of CH_3COOH = 4.7)

(1) 1 (2) 4.7

(3) 7 (4) 14

74. If solubility of weak salt A_2X_3 is s $molL^{-1}$ then its K_{sp} is

(1) s^2 (2) $4s^3$

(3) $9s^4$ (4) $108s^5$

75. Molar solubility of $Ni(OH)_2$ in 0.5 M $Ca(OH)_2$ is (K_{sp} of $Ni(OH)_2 = 2 \times 10^{-15}$ M)

(1) 2×10^{-13} M (2) 2×10^{-14} M

(3) 2×10^{-15} M (4) 2×10^{-16} M

76. For the reaction $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ equilibrium concentrations of SO_2, O_2, and SO_3 are 0.1 M, 0.2 M and 0.1 M respectively. The value of K_C for the reaction is

(1) 0.1 (2) 1

(3) 2 (4) 5

77. Which salt undergoes cationic hydrolysis

(1) NaCl (2) NH_4Br

(3) CH_3COONa (4) NaBr

78. Which of the following is not an oxidation?

(1) Addition of electronegative element to a substance

(2) Removal of electropositive element from a substance

(3) Loss of electron(s) by any species

(4) Decrease in oxidation number of an element

79. Blue colour starts developing when

(1) Zn rod is dipped in copper nitrate solution

(2) Cu rod dipped in silver nitrate solution

(3) Zn rod dipped in silver nitrate solution

(4) Cu and dipped in zinc nitrate solution

80. Oxidation number of oxygen in superoxide is

(1) 0 (2) –1

(3) $-\frac{1}{2}$ (4) –2

81. Combination reaction among the following is

(1) $2KClO_3 \rightarrow 2KCl + 3O_2$

(2) $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

(3) $Cr_2O_3 + 2Al \rightarrow Al_2O_3 + 2Cr$

(4) $Mg + 2H_2O \rightarrow Mg(OH)_2 + H_2$

82. Oxidation number of middle Br in Br_3O_8 is

(1) 0 (2) –1

(3) 4 (4) 6

83. $xCr_2O_7^{2-} + ySO_3^{2-} + zH^+ \rightarrow 2xCr^{3+} + ySO_4^{2-} + \frac{z}{2}H_2O$

The value of x and y in above balanced equation are respectively

(1) 1, 3 (2) 3, 1

(3) 2, 3 (4) 3, 2

84. Which of the following is a self indicator?

(1) MnO_4^- (2) $Cr_2O_7^{2-}$

(3) I^- (4) $S_2O_3^{2-}$

85. Given the standard electrode potentials

$K^+/K = -2.93$ V, $Ag^+/Ag = 0.80$ V

$Hg^{2+}/Hg = 0.79$ V, $Mg^{2+}/Mg = -2.37$ V

The element having strongest reducing power is

(1) K (2) Mg

(3) Ag (4) Hg

86. Oxidation number of P in $H_4P_2O_7$ is

(1) 0 (2) –3

(3) +3 (4) +5

87. Which among the following can not show disproportionation?

(1) PH_3 (2) H_2O_2

(3) S_8 (4) $HClO_2$

88. Example of redox reaction is

(1) $Na + HCl \rightarrow NaCl + H_2$

(2) $Na_2O + 2HCl \rightarrow 2NaCl + H_2O$

(3) $NaOH + HCl \rightarrow NaCl + H_2O$

(4) $Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$

89. Fractional oxidation number is observed in

(1) $Na_2S_4O_6$ (2) CH_3COOH

(3) Fe_2O_3 (4) O_2F_2

90. Oxidation number of hydrogen in CaH_2 is

(1) 0 (2) +1

(3) –1 (4) –2

Botany

91. All of the following are considered to be the event concerned with cell cycle, **except**

(1) DNA replication

(2) Glucose synthesis

(3) Cytoplasmic increase

(4) Genome duplication

92. The approximate time taken to divide once by a typical human cell in culture is

(1) 24 hours (2) 38 minutes

(3) 90 minutes (4) 30 hours

93. The time taken by a yeast cell to complete its cell cycle is

(1) Less than the time taken by a human cell

(2) 12 hours

(3) Same as the time taken by a typical human cell

(4) 18 – 20 hours

94. If a human cell takes X unit time to complete its cell cycle then, what would be the duration of its M phase?

(1) More than 95% of X (2) X/2 unit time

(3) 95% of X (4) Less than 5% of X

95. M phase of a cell ends with

(1) Separation of chromosomes

(2) Replication of DNA

(3) Division of cytoplasm

(4) Condensation of chromatin

96. The interphase of cell cycle includes all of the following phases, **except**

(1) G_2 phase (2) S phase

(3) G_1 phase (4) G_0 phase

97. Which of the following sequences regarding different phases of a cell is acceptable?

(1) $M \rightarrow G_1 \rightarrow G_0 \rightarrow G_1$ (2) $G_1 \rightarrow G_2 \rightarrow S \rightarrow M$

(3) $M \rightarrow S \rightarrow G_1 \rightarrow G_0$ (4) $G_0 \rightarrow G_1 \rightarrow G_2 \rightarrow S$

98. Doubling of the amount of DNA in a cell occurs during

(1) The phase that occurs just after M phase

(2) Synthesis phase

(3) G_2 phase

(4) The phase that occurs just after G_2 phase

99. Select the **incorrect** statement from the following.

(1) In human, mitotic cell division is only seen in the diploid somatic cell

(2) The cells in quiescent stage remain metabolically active

(3) In male honey bee, haploid cells do not divide by mitosis

(4) Plants can show mitotic division in both haploid and diploid cells

100. Mitotic division is also called equational division because

(1) Parent cell is divided into two daughter cells

(2) Number of chromosomes in the parent and progeny cells is the same

(3) Size of parent cell and daughter cells is the same

(4) The daughter cells have just half of the number of chromosomes as that of parent cell

101. Prophase is marked by the

(1) Initiation of condensation of chromosomal material

(2) Duplication of centrosome

(3) Duplication of cell organelles

(4) Development of nuclear envelope

102. In prophase, each chromosome has two chromatids. These chromatids are attached together at the

(1) Telomere (2) Centriole

(3) Kinetochore (4) Centromere

103. Mitotic apparatus is formed of

(1) Two centrosomes with chromatids

(2) Two asters together with spindle fibres

(3) Two chromatids with acrosome

(4) One primary constriction with two chromatids

104. The beginning of second phase of mitosis is marked by

(1) Complete disintegration of the nuclear envelope

(2) Attachment of spindle fibres with kinetochore

(3) Formation of nucleolus

(4) Formation of spindle fibres and asters

105. In which phase of cell cycle, chromosomes move to spindle equator?

(1) G_2 phase (2) Prophase

(3) Metaphase (4) Anaphase

106. Which of the following is **not** a characteristic of anaphase?

(1) Movement of chromosomes away from the equatorial plate

(2) Condensation of chromosomes

(3) Splitting of centromere

(4) Separation of chromatids of a chromosome

107. Chromosomes decondense and lose their individuality during

(1) Anaphase (2) Metaphase

(3) Telophase (4) Interphase

108. Cytokinesis is

(1) Formation of cytoplasmic organelles

(2) Synthesis of cytoplasmic material

(3) Division of cell into daughter cells by separation of cytoplasm

(4) Disappearance of membrane bound organelles in the cytoplasm

109. During cell division in plants, the formation of new cell wall begins with the formation of a simple precursor called

(1) Middle lamella (2) Cell plate

(3) Microtubule (4) Axoneme

110. Which of the following leads to the formation of syncytium?

(1) Nuclear division without DNA duplication

(2) Nuclear division without the disintegration of nuclear membrane

(3) Cell wall formation before the nuclear division

(4) Karyokinesis not followed by cytokinesis

111. Meiosis involves

(1) Single cycle of nuclear division

(2) Single cycle of DNA replication

(3) Single cycle of cytoplasmic division

(4) Two cycle of DNA replication

112. Select the **incorrect** match.

(1) Leptotene – Compaction of chromosome

(2) Pachytene – Bivalents clearly appears as tetrads

(3) Zygotene – Dissolution of synaptonemal complex

(4) Diplotene – Appearance of chiasmata

113. In which stage of prophase I, recombinase mediated process is involved?

(1) Pachytene (2) Zygotene

(3) Diakinesis (4) Leptotene

114. The stage of cell division in which terminalisation of chiasmata occurs is

(1) The final stage of prophase I

(2) The first stage of meiosis II

(3) Metaphase I

(4) The stage between pachytene and diplotene

115. Metaphase I is different from metaphase II as in the former

(1) Bivalent chromosomes align on the equatorial plate

(2) Homologous chromosomes separate from each other

(3) Crossing over between non-sister chromatids of homologous chromosomes occur

(4) Metaphasic plate is not formed

116. The condition which is called as dyad of cells is seen during

(1) Anaphase I (2) Anaphase II

(3) Metaphase II (4) Telophase I

117. Microtubules from opposite poles of the spindle get attached to the kinetochores of a chromosome in

(1) Prophase I of meiosis and prophase of mitosis

(2) Metaphase I and metaphase II

(3) Metaphase of mitosis and metaphase II

(4) Prophase of mitosis and metaphase I

118. The stage of meiotic division called interkinesis is followed by

(1) Anaphase I (2) Telophase I

(3) Prophase II (4) Meiosis I

119. Though, the union of gametes results in doubling of chromosome number, which process conserve specific chromosome number in such organisms?

(1) Mitosis (2) Meiosis

(3) Crossing over (4) Recombination

120. A group of ten diploid male gamete mother cells are going to form male gametes. How many cells of stage metaphase II will be seen during this process?

(1) Ten (2) Twenty

(3) Forty (4) Eighty

121. Which one of the following is the significance of meiosis?

(1) Production of identical daughter cells

(2) Cell repair

(3) Increase in the genetic variability in the population

(4) Continuous growth of plant body

122. Longer distance transport of substance in plants through xylem and phloem is called

(1) Transfusion (2) Diffusion

(3) Ascent of sap (4) Translocation

123. Read the following statements and choose the option which is true for them.

Statement-A: Diffusion in solids is more likely rather than of solids

Statement-B: Diffusion rates are affected by the gradient of concentration.

(1) Both the statements are true

(2) Both the statements are false

(3) Only statement A is true

(4) Only statement B is true

124. Regarding active transport, which one of the following is **not** correct?

(1) Energy is used in this process

(2) It occurs against the concentration gradient

(3) It is carried out by specific membrane-proteins

(4) The pump molecules are not sensitive to inhibitors

125. In plant cells, water channels are made up of _____ of aquaporins.

Select the **correct** option to fill in the blank.

(1) Four same type

(2) Eight different types

(3) Four different types

(4) Eight same type

126. In passive symport

(1) Two types of molecules cross the membrane in same direction

(2) Two types of molecules cross the membrane in opposite direction

(3) Only one molecule can cross the membrane

(4) Same type of molecules cross the membrane simultaneously but in opposite direction

127. Which of the following mechanism is/are highly selective in nature?

(a) Facilitated transport

(b) Simple diffusion

(c) Active transport

(1) (b) & (c) (2) (a) & (c)

(3) (a) & (b) (4) (a) only

128. Water potential of a solution increases when

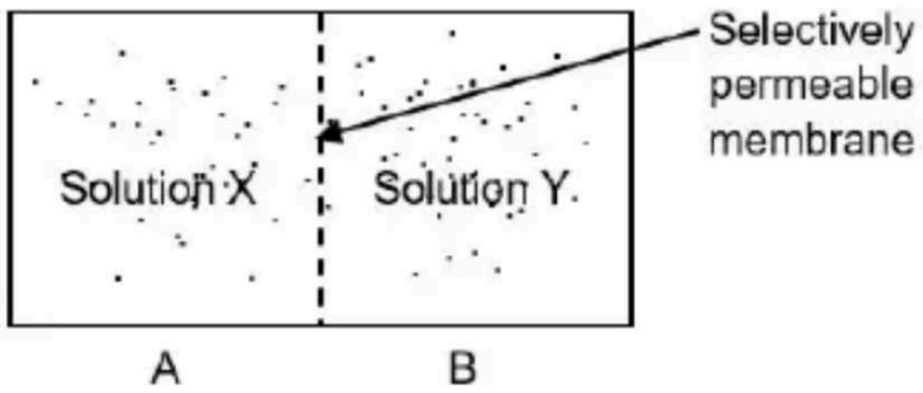
(1) More solute is dissolved in the solution

(2) Water molecule is removed from the solution

(3) Pressure greater than the atmospheric pressure is applied on the solution

(4) Temperature of the solution is decreased

129. Which values for pressure potential and solute potential at standard temperature will favour the movement of water from chamber B to chamber A?



(1) ψ_s (X) = -6 bar ; ψ_p (X) = 4 bar

ψ_s (Y) = -4 bar ; ψ_p (Y) = 3 bar

(2) ψ_s (X) = -8 bar ; ψ_p (X) = 7 bar

ψ_s (Y) = -3 bar ; ψ_p (Y) = 1 bar

(3) ψ_s (X) = -5 bar ; ψ_p (X) = 3 bar

ψ_s (Y) = -5 bar ; ψ_p (Y) = 2 bar

(4) ψ_s (X) = -6 bar ; ψ_p (X) = 5 bar

ψ_s (Y) = -7 bar ; ψ_p (Y) = 2 bar

130. When a cell is placed in hypotonic solution water diffuses (i) the cell causing the cytoplasm build up a pressure against the wall, that is called (ii) .

Select the correct option to fill in the blanks (i) and (ii).

(1) (i) – Out of ; (ii) – Wall pressure

(2) (i) – Out of ; (ii) – Osmotic pressure

(3) (i) – Into ; (ii) – Turgor pressure

(4) (i) – Into ; (ii) – Wall pressure

131. In imbibition, water movement

(1) Is against the concentration gradient

(2) Is along the concentration gradient

(3) Requires energy of ATP

(4) Requires special transport proteins

132. The girdling experiment in plants shows that

(1) Xylem is also responsible for the transport of food

(2) Phloem can transport both carbohydrates and minerals

(3) The translocation of water in xylem is unidirectional

(4) Phloem is responsible for translocation of food towards root

133. The substances that are transported through xylem are

(1) Only inorganic nutrients

(2) Only water and minerals

(3) Many organic compounds except amino acid

(4) Both organic and inorganic materials

134. Most mineral elements from the soil enter the root epidermal cells

(1) In the form of atoms

(2) By active absorption

(3) By osmosis

(4) By simple diffusion

135. In plants symplastic pathway does **not** involve

(1) Intercellular space

(2) Cytoplasm

(3) Plasma membrane

(4) Plasmodesmata

Zoology

136. Functions of different organ systems in our body must be coordinated to maintain homeostasis with the help of

(1) Only sensory system

(2) Only endocrine system

(3) Respiratory system

(4) Jointly by neural system and endocrine system

137. Which of the following functions is **not** included in neural system?

(1) Transmission of nerve impulse

(2) Impulse conduction across a synapse

(3) Reflex action

(4) Chemical integration only through hormones

138. Select the **incorrect** statement w.r.t. functions of neurons

(1) Generate electrical signals

(2) Receive stimuli

(3) Transmit impulse

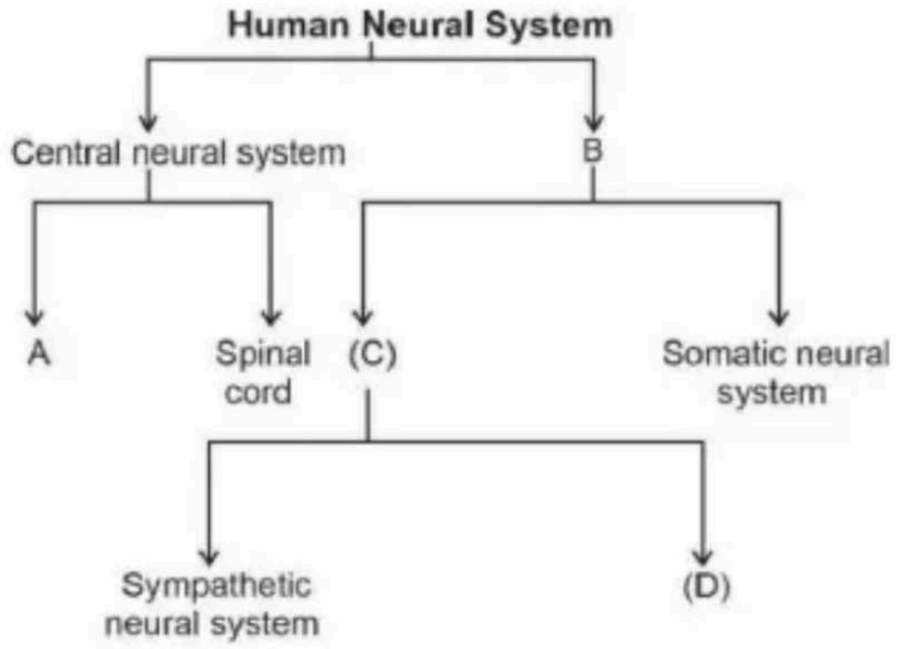
(4) Produce different kinds of stimuli

139. Simplest neural system composed of network of neurons is present in

(1) *Hydra* (2) *Fasciola*

(3) *Pheretima* (4) *Periplaneta*

140. On the basis of the following given flow chart w.r.t. human neural system, Identify A, B, C and D. Choose the option which represents A, B, C, D **correctly**



	A	B	C	D
1.	Ganglia	Peripheral neural system	Parasympathetic neural system	Brain
2.	Brain	Parasympathetic neural system	Peripheral neural system	Ganglia
3.	Brain	Peripheral neural system	Autonomic neural system	Parasympathetic neural system
4.	Peripheral neural system	Brain	Parasympathetic neural system	Ganglia

141. All of the following are included in visceral nervous system **except**

(1) Nerve plexuses

(2) Ganglia

(3) Sensory and motor neurons

(4) Spinal cord

142. Which of the following is **incorrect** w.r.t. synaptic knob?

(1) Bulb like structure at axon terminal

(2) Contains neurotransmitters

(3) Possess synaptic vesicles

(4) Contains Nissl's granules in cytoplasm

143. Neuron with one axon and two or more dendrites is called

(1) Pseudounipolar (2) Bipolar

(3) Multipolar (4) Unipolar

144. Select the **incorrect** statement w.r.t. unmyelinated nerve fibres in Peripheral neural system (PNS)

(1) Enclosed by Schwann cells around axon.

(2) Absence of myelin sheath around axon

(3) Found in somatic neural system

(4) Not found in autonomous neural system

145. Neurons are excitable cells because their membranes are in

(1) Polarised state

(2) Depolarised state

(3) Repolarised state

(4) Hyperpolarised state

146. When a neuron is not conducting any impulse in its resting condition, the axonal membrane is comparatively more permeable to

(1) Na^+ (2) K^+

(3) Cl^- (4) Ca^{+2}

147. Ionic gradients across the resting membrane are maintained by

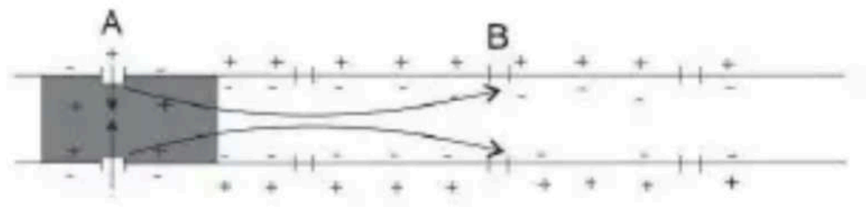
(1) Rapid influx of Na^+ down its concentration gradient

(2) Rapid efflux of K^+ down its concentration gradient

(3) Active transport of ions by $Na^+ – K^+$ pump

(4) Leakage of more K^+ than Na^+ against concentration gradient

148. In the following diagrammatic representation of an axon, stimulus is applied at point A and action potential is generated. When impulse reaches at point B, what will be the direction of current?



(1) A to B on outer surface of axon

(2) B to A on outer surface of axon

(3) B to A on inner surface of axon

(4) A to B on both outer and inner surface of axon

149. At the site of excitation, resting potential of axonal membrane within a fraction of seconds is restored mainly by

(1) Ca^{+2} pump

(2) Diffusion of K^+ inside axoplasm

(3) Rapid diffusion of K^+ outside the membrane

(4) Rapid diffusion of Na^+ inside axon

150. Neuronal junctions are called

(1) Synapse

(2) Joints

(3) Desmosomes

(4) Tight junction

151. Transmission of impulse across electrical synapse is very similar to impulse conduction along

(1) Neuromuscular junction

(2) A chemical synapse in CNS

(3) A chemical synapse in PNS

(4) A single axon

152. A synapse is formed by

(1) Only pre-synaptic membrane

(2) Only post-synaptic membrane

(3) Membranes of a pre-synaptic neuron and a post-synaptic neuron, always separated by a synaptic cleft

(4) Membranes of a pre-synaptic neuron and a post-synaptic neuron, which may or may not be separated by a gap

153. All of the following characteristics of electrical and chemical synapses **correctly** differentiate these synapses **except**

	Electrical Synapse	Chemical Synapse
1.	Electrical current can flow directly from one neuron into the other	Electrical current cannot flow directly from one neuron into the other
2.	Pre-synaptic and post- synaptic neurons are in close proximity	Pre-synaptic and post synaptic neurons are comparatively not in close proximity
3.	Impulse transmission across synapse is always slower	Impulse transmission across synapse is always faster
4.	They are rare in our body	They are more abundant in our body

154. Synaptic vesicles move towards synaptic cleft where they fuse with plasma membrane, when

(1) An action potential arrives at the axon terminal of pre-synaptic neuron

(2) An impulse is generated in post-synaptic membrane

(3) Neurotransmitters bind with pre-synaptic membrane

(4) An action potential is generated in cyton

155. Read the following statements carefully

(a) Neurotransmitters bind to their specific receptors present on post-synaptic membrane

(b) Binding of neurotransmitters open ion channels in post-synaptic membrane

(c) Entry of ions in post-synaptic neuron always generates excitatory post-synaptic potential

(d) Entry of ions in post-synaptic neuron always generates inhibitory post-synaptic potential

Choose the option with all **correct** statements

(1) a and b (2) b and c

(3) c and d (4) a and d

156. Arrange the cranial meninges in a **correct** sequence starting from the meninx invested on cerebrum

(1) Dura mater → arachnoid → pia mater

(2) Pia mater → arachnoid → dura mater

(3) Arachnoid → pia mater → dura mater

(4) Pia mater → dura mater→ arachnoid

157. Which of the following structure of brain is **not** included in forebrain?

(1) Cerebrum (2) Cerebral aqueduct

(3) Thalamus (4) Hypothalamus

158. Corpus callosum is a tract of nerve fibres which connects

(1) Right and left cerebral hemispheres

(2) Two cerebellar hemispheres

(3) Forebrain to midbrain

(4) Cerebrum to mid brain

159. All of the following centres are situated in cerebral cortex **except**

(1) Sensory areas

(2) Motor areas

(3) Association areas

(4) Major coordinating centre for sensory and motor signaling

160. Which part of hindbrain is **not** a part of brain stem?

(1) Cerebrum (2) Pons

(3) Midbrain (4) Cerebellum

161. The area of cerebral cortex which is associated with complex functions like intersensory associations, memory and communication is

(1) Sensory area

(2) Motor area

(3) Association Area

(4) Thermoregulatory area

162. Cerebrum wraps around a structure of forebrain called

(1) Corpus callosum

(2) Corpora quadrigemina

(3) Thalamus

(4) Pons

163. Group of neurosecretory cells present in hypothalamus are mainly responsible for

(1) Control of body temperature

(2) Secretion of hormones

(3) Urge of eating and drinking

(4) Swallowing

164. Limbic system does **not** regulates

(1) Sexual behaviour

(2) Expression of emotional reactions

(3) Motivation

(4) Gastric secretions

165. Cardiovascular reflexes are under the control of

(1) Medulla oblongata (2) Pons

(3) Cerebrum (4) Cerebellum

166. Which part of our CNS acts as a master clock?

(1) Pituitary gland

(2) Pineal gland

(3) Cerebrum

(4) Hypothalamus

167. Immediate and involuntary response to a peripheral nervous stimulation with the help of CNS without conscious effort is called

(1) Peripheral response

(2) Voluntary action

(3) Reflex action

(4) Rhythm of the body

168. Which of the following neuron is **not** present in reflex arc of knee-jerk reflex?

(1) Excitatory neuron (2) Receptor neuron

(3) Effector neuron (4) Interneuron

169. Read the following statements **A** and **B** and choose the **correct** option

(A) Changes in the environment are detected by sensory organs

(B) Sensory organs send appropriate signals to CNS where all inputs are processed and analysed

(1) Only Statement **A** is correct

(2) Only statement **B** is incorrect

(3) Both statements **A** and **B** are incorrect

(4) Both statements **A** and **B** are correct

170. Choose the **incorrect** statement from the given below statements

(1) Mucus coated olfactory epithelium consists of three kinds of cells

(2) Sensory neurons of the olfactory epithelium terminates into extensions of brain's limbic system

(3) The chemical senses of taste and smell are functionally dissimilar and interrelated

(4) The tongue detects tastes through taste buds, containing gustatory receptors.

171. Thick anterior part of choroid is called

(1) Cornea (2) Ciliary body

(3) Lens (4) Pupil

172. Select the **mismatch** w.r.t. structure of human eye

(1) Aqueous chamber – space between cornea and lens

(2) Vitreous chamber – space between lens and retina

(3) Cornea – Anterior portion of iris

(4) Choroid – Middle layer of eye, bluish in colour

173. All of the following properties of rods and cones in the given Column-I and Column-II differentiate them correctly **except**

	Column-I	Column-II
1.	Rods are responsible for scotopic vision	Cones are responsible for photopic vision
2.	Visual pigment is rhodopsin or visual purple in rods	Presence of three types of visual pigments in cones
3.	Rods are responsible for twilight vision	Cones are responsible for daylight vision
4.	When rods are stimulated equally a sensation of white light is produced	When cones are stimulated equally, a sensation of violet colour light is produced

174. Greatest power of resolution in human's eye is found in the region called

(1) Yellow spot (2) Blind spot

(3) Fovea (4) Macula

175. Read the following statements w.r.t. mechanism of vision and choose the option which **correctly** represents true and false statements

A. Ultraviolet rays focussed on the retina generate impulses in photoreceptor cells

B. Changes in structure of retinal causes membrane permeability changes in rods and cones

C. Signal produced in rods and cones generates action potential in ganglionic cells through the bipolar cells.

D. Image formed on the retina is recognised by visual cortex based on earlier memory and experience

	A	B	C	D
1.	True	False	True	False
2.	False	False	True	True
3.	False	True	False	True
4.	True	True	True	False

176. The middle ear contains three ossicles in chain like fashion in a order from inside to outside as

(1) Malleus, incus, stapes

(2) Incus, stapes, malleus

(3) Stapes, incus, malleus

(4) Incus, malleus, stapes

177. The eustachian tube helps in equalising the pressure on either sides of the

(1) Ear drum

(2) Oval window

(3) Round window

(4) Bony labyrinth and membranous labyrinths

178. Perilymph is **not** found in

(1) Scala vestibuli

(2) Scala tympani

(3) Scala media

(4) In between bony and membranous labyrinths

179. All of the following structures have sensory hair cells **except**

(1) Macula

(2) Crista

(3) Organ of corti

(4) Reissner's membrane

180. Read following statements carefully and choose the option with all **incorrect** statements

(a) The vestibular apparatus is located below the cochlea

(b) The bony canals are suspended in the endolymph of membranous canals

(c) Ear drum vibrates in response to the vibrations in perilymph

(d) The waves in the lymphs induce a ripples in the basilar membrane

(1) a and d (2) a, c and d

(3) b, c and d (4) a, b and c

❑❑❑

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NCERT BOOSTER TEST SERIES

Test – 03 (Answer Key)

1.	(4)	46.	(1)	91.	(2)	136.	(4)
2.	(1)	47.	(2)	92.	(1)	137.	(4)
3.	(4)	48.	(3)	93.	(1)	138.	(4)
4.	(4)	49.	(3)	94.	(4)	139.	(1)
5.	(1)	50.	(3)	95.	(3)	140.	(3)
6.	(1)	51.	(2)	96.	(4)	141.	(4)
7.	(3)	52.	(2)	97.	(1)	142.	(4)
8.	(4)	53.	(2)	98.	(2)	143.	(3)
9.	(2)	54.	(4)	99.	(3)	144.	(4)
10.	(3)	55.	(3)	100.	(2)	145.	(1)
11.	(1)	56.	(2)	101.	(1)	146.	(2)
12.	(3)	57.	(3)	102.	(4)	147.	(3)
13.	(4)	58.	(4)	103.	(2)	148.	(2)
14.	(2)	59.	(1)	104.	(1)	149.	(3)
15.	(2)	60.	(2)	105.	(3)	150.	(1)
16.	(2)	61.	(4)	106.	(2)	151.	(4)
17.	(2)	62.	(3)	107.	(3)	152.	(4)
18.	(1)	63.	(3)	108.	(3)	153.	(3)
19.	(3)	64.	(4)	109.	(2)	154.	(1)
20.	(3)	65.	(3)	110.	(4)	155.	(1)
21.	(2)	66.	(4)	111.	(2)	156.	(2)
22.	(3)	67.	(2)	112.	(3)	157.	(2)
23.	(3)	68.	(1)	113.	(1)	158.	(1)
24.	(3)	69.	(2)	114.	(1)	159.	(4)
25.	(2)	70.	(1)	115.	(1)	160.	(4)
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